



AI-POWERED REVIEW ANALYSIS PLATFORM

Using HuggingFace + AWS SageMaker



✓ TASK 0: PREPROCESSING – MAPPING STAR RATINGS TO SENTIMENT CLASSES

Goal: Convert star ratings (1 to 5) into sentiment classes

Rating.	Sentiment
1 - 2.	Negative 😞
3.	Neutral 😐
4 - 5.	Positive 😊

- Cleaned and filtered review dataset
- Mapped star ratings to sentiment categories
- Saved as: amazon_reviews_with_sentiment.csv



TASK 1: CLASSIFICATION USING PRE-TRAINED TRANSFORMERS

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Objective: Classify reviews as Positive, Neutral, or Negative

Steps Taken:

- ✓ Selected distilbert-base-uncased (lightweight and fast)
- ✓ Tokenized and fine-tuned on custom dataset
- ✓ Trained on 1000 reviews for fast iteration
- ✓ Evaluated using `classification_report` and `confusion_matrix`
- ✓ Saved Fine-Tuned Model
 - Used `model.save_pretrained()`
 - Uploaded to S3 Bucket
 - Example file: `model.tar.gz`

Result:

- Great accuracy on validation data
- Clear classification between all 3 classes

TASK 2: PRODUCT CATEGORY CLUSTERING

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Objective: Cluster reviews into 4–6 major product categories

Process:

-  Used SentenceTransformer (MiniLM-L6-v2) to generate embeddings
-  Applied KMeans clustering with k=5
-  Visualized using t-SNE
-  Saved the output: clustered_reviews.csv

Example Clusters:

- eBook Readers
- Accessories
- Kitchen items
- Pet gear
- Miscellaneous electronics

 Each review now includes a cluster number!

TASK 3: SUMMARIZATION WITH GENERATIVE AI

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Objective: Generate blog-style summaries per cluster

🤖 Used:

- HuggingFace facebook/bart-large-cnn for summarization
- Collected representative reviews from each cluster
- Generated clear summaries with:
 - ♦ Top 3 products
 - ♦ Common complaints
 - ♦ The worst product and why to avoid it

📄 Output: Category-based summaries for quick product comparison!

TASK 4: DEPLOYMENT ON AWS SAGEMAKER

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Objective: Deploy the fine-tuned model for real-time predictions

 **Deployment Steps:**

-  Saved model and tokenizer
-  Compressed model folder using tar.gz
-  Uploaded to S3 bucket
-  Created SageMaker Model
-  Deployed via endpoint: sentiment-endpoint
-  Confirmed InService status





FINAL THOUGHTS

- The platform handles full ML pipeline:
data → model → clustering → summaries
→ cloud deployment
- Easily expandable with more data or
multilingual support
- Ready for real-world integration (e.g., e-
commerce platforms)



THANK YOU!

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